# Calculus & Analytical Geometry-I

## MIDTERM SOLVED PAPERS (PAPER #1)

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**Question No: 1 ( Marks: 1 ) - Please choose one**

If f is a twice differentiable function at a stationary point $x_0$ and $f''(x_0) > 0$ then f has relative ……. At $x_0$

- Minima
- Maxima
- None of these

**Question No: 2 ( Marks: 1 ) - Please choose one**

If f is a twice differentiable function at a stationary point $x_0$ and $f''(x_0) < 0$ then f has relative ……. At $x_0$

- Minima
- Maxima
- None of these

**Question No: 3 ( Marks: 1 ) - Please choose one**

A line $y = y_0$ is called a ……. for the graph f if

$$\lim_{x \to +\infty} f(x) = y_0 \quad or \quad \lim_{x \to -\infty} f(x) = y_0$$

- Vertical asymptotes
- Horizontal asymptotes
- None of these

**Question No: 4 ( Marks: 1 ) - Please choose one**

If $f(x) = 3x^8 + 2x + 1$ then $f'(x) =$ _____

- $3x^7 + 2$
- $24x^7 + 2$
- $3x^9 + 2x^2$
- $24x^9 + 2x^2$

**Question No: 5 ( Marks: 1 ) - Please choose one**

$\frac{d(\tan x)}{dx} =$

- $\sec x$
- $\sec^2 x$
- $co\sec x$
- $co\sec^2 x$

**Question No: 6 ( Marks: 1 ) - Please choose one**

If $xy = 4$ then $\frac{dy}{dx} =$

- 0
- $\frac{-1}{x^2}$
- $\frac{4}{x^2}$
- $\frac{-4}{x^2}$

**Question No: 7 ( Marks: 1 ) - Please choose one**

If $2x - y = -3$ then $\frac{dy}{dx} =$

- 2
- -2
- 0
- -3

**Question No: 8 ( Marks: 1 ) - Please choose one**

$\frac{d}{dx}[\sec x] =$ _____

- $\frac{1}{1+\sin^2 x}$
- $\frac{-\sin x}{1+\sin^2 x}$
- $\frac{1}{1-\sin^2 x}$
- $\frac{\sin x}{1-\sin^2 x}$

**Question No: 9 ( Marks: 1 ) - Please choose one**

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$30^0$ = _____

- $\frac{\pi}{3}$
- $\frac{\pi}{4}$
- $\frac{\pi}{6}$
- $\frac{\pi}{2}$

**Question No: 10 ( Marks: 1 ) - Please choose one**

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Consider a function $h(x)$ and a constant $c$ then
$\frac{d}{dx}\left((c)\{h(x)\}\right)$ = _____

- 0
- $\frac{d}{dx}\left(h(x)\right)$
- $\frac{d}{dx}\left(h(cx)\right)$
- $c\frac{d}{dx}\left(h(x)\right)$

**Question No: 11 ( Marks: 1 ) - Please choose one**

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Suppose that $f$ and $g$ are differentiable functions of $x$ then
$\frac{d}{dx}\left[\frac{f}{g}\right]$ =

- $\frac{[g][f']-[f][g']}{g^2}$
- $\frac{[g'][f]-[f'][g]}{g^2}$

$\frac{[g][f']-[f][g']}{f^2}$

3

$\frac{[g'][f]-[f'][g]}{f^2}$

3

**Question No: 12 ( Marks: 1 ) - Please choose one**

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$\frac{d}{dx}[\text{cosec } x] =$ _____

$\frac{1}{1+\cos^2 x}$

3

$\frac{-\cos x}{1-\cos^2 x}$

3

$\frac{-\cos x}{1-\cos^2 x}$

3

$\frac{1}{1-\cos^2 x}$

3

**Question No: 13 ( Marks: 1 ) - Please choose one**

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$y = f\left(g\left(h(x)\right)\right)$

If

$u = g\left(h(x)\right)$

$v = h(x)$ then $\frac{dy}{dx}$ = _____

$\frac{dy}{du}.\frac{du}{dv}.\frac{dv}{dx}$

3

$\frac{dy}{du}\frac{du}{dv}\frac{dv}{dx}$

3

$\frac{dv}{du}.\frac{du}{dv}.\frac{dy}{dx}$

3

**Question No: 14 ( Marks: 1 ) - Please choose one**

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Chain rule is a rule for differentiating _____ of functions.

3 Product

3 Sum

- Difference
- Composition

**Question No: 15 ( Marks: 1 ) - Please choose one**

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The power rule, $\frac{d}{dx}[x^n] = nx^{n-1}$ holds if n is _____

- An integer
- A rational number
- An irrational number
- All of the above

**Question No: 16 ( Marks: 1 ) - Please choose one**

Let a function $f$ be defined on an interval, and let $x_1$ and $x_2$ denotes two distinct points in that interval. If $f(x_1) = f(x_2)$ for all points $x_1$ and $x_2$ then which of the following statement is correct?

- $f$ is a decreasing function
- $f$ is an increasing function
- $f$ is a constant function

**Question No: 17 ( Marks: 1 ) - Please choose one**

If $f''(x) < 0$ on an open interval (a,b) then which of the following statement is correct?

- $f$ is concave up on (a, b).
- $f$ is concave down on (a, b)
- $f$ is linear on (a, b).

**Question No: 18 ( Marks: 1 ) - Please choose one**

*If* $x > \frac{1}{2}$ *then* $\frac{d}{dx}[\ln 2x] =$ _____

- $\frac{1}{x}$
- $\frac{2}{x}$
- $\frac{1}{2x}$
- $\frac{2}{x^2}$

**Question No: 19 ( Marks: 1 ) - Please choose one**

Let $y = (x^3 + 2x)^{37}$. Which of the following is correct?

$\frac{dy}{dx} = (37)(x^3 + 2x)^{36}$

3

$\frac{dy}{dx} = 111x^2(x^3 + 2x)^{36}$

3

$\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{36}$

3

$\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{38}$

3

**Question No: 20 ( Marks: 1 ) - Please choose one**

If we have $x^2 + y^2 = 1$ then $\frac{dy}{dx} =$ _____

$\frac{-x}{y}$

3

$\frac{x}{y}$

3

$\frac{-y}{x}$

3

3 None of these

**Question No: 21 ( Marks: 1 ) - Please choose one**

$\log_b ac =$ _____

$\log_b a + \log_b c$

3

$\log_b a - \log_b c$

3

$\frac{\log_b a}{\log_b c}$

3

$(\log_b a)(\log_b c)$

3

**Question No: 22 ( Marks: 1 ) - Please choose one**

$\log_b a^r =$ _____

- $a \log_b r$
- $r \log_b a$
- $\frac{\log_b a}{\log_b r}$
- $\log_b a + \log_b r$

**Question No: 23 ( Marks: 1 ) - Please choose one**

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$\log_b \frac{1}{c} =$ _____

- $\log_b c$
- $1 - \log_b c$
- $-\log_b c$
- $1 + \log_b c$

**Question No: 24 ( Marks: 1 ) - Please choose one**

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$\log_b \frac{1}{t} =$ _____

- $\log_b t$
- $1 - \log_b t$
- $1 + \log_b t$
- $-\log_b t$

**Question No: 25 ( Marks: 3 )**

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Differentiate:

$$f(t) = \left(t^3 + 4\right)^4$$

$$f'(t) = 4\left(t^3 + 4\right)^3 . \frac{d}{dx}\left(t^3 + 4\right)$$

$$f'(t) = 4\left(t^3 + 4\right)^3 . 3t^2$$

$$f'(t) = 12t^2\left(t^3 + 4\right)^3$$

**Question No: 26 ( Marks: 5 )**

Differentiate $\sqrt{13x^2-5x+8}$

$$f`(x)=\frac{1}{2\sqrt{13x^2-5x+8}}\frac{d}{dx}13x^2-5x+8$$

$$f`(x)=\frac{1}{2\sqrt{13x^2-5x+8}}.26x-5$$

**Question No: 27 ( Marks: 10 )**

Differentiate the following function

$$f(x)=x^3\,.\,e^{\frac{1}{x}}$$

$$f`(x)=3x^2\,.\,e^{\frac{1}{x}}+x^3\,.\,e^{\frac{1}{x}}-\frac{1}{x^2}$$

$$f`(x)=3x^2\,.\,e^{\frac{1}{x}}-\frac{x^3\,.\,e^{\frac{1}{x}}}{x^2}$$

$$f`(x)=e^{\frac{1}{x}}[3x^2-\frac{x^3}{x^2}]$$

$$f`(x)=e^{\frac{1}{x}}[\frac{3x^4-x^3}{x^2}]$$

$$f`(x)=xe^{\frac{1}{x}}[\frac{3x^3}{x^2}-\frac{x^2}{x^2}]$$

$$f`(x)=xe^{\frac{1}{x}}[3x-1]Ans$$

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